

ngram → orden

Document	Text
D1	students watch youtube
D2	youtube <u>watch</u> youtube
D3	students write comments
D4	youtube write comments

students	watch	youtube	write	comments
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bigram

1 students watch / watch youtube / youtube watch
2
3
4 student write / write comments / comments write
5
6

	student watch	watch youtube	youtube watch	student write	write comments	youtube write
D1	1	1	0	0	0	0
D2	0	0	0	0	1	0
D3	0	0	0	1	0	0
D4	0	0	0	0	0	1

0210					

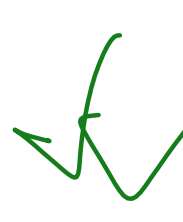
Semantic

Goodmontage

This movie is good

This movie is not good

N-grams solve some problems of Bag of Words, but they introduce new problems.



1. Vocabulary size explodes

Suppose your vocabulary has 10,000 unique words.

Unigrams (BoW)

dog
cat
eat
food
...

Vocabulary size $\approx$ 10,000

Bigrams

dog eats
eats food
cat drinks
...

Possible combinations become much larger.

✓ Trigrams

the dog eats
dog eats food
...

Vocabulary can grow to **hundreds of thousands or millions** of features.

More memory required
More computation required

2. Sparse vectors become worse

Most n-grams appear very rarely.

Training data

I love machine learning
I love deep learning

Vocabulary:

I love
love machine
machine learning
love deep
deep learning

New sentence

I love NLP

Many features become:
[1,0,0,0,0]

Lots of zeros.

This is called **sparsity**.

Harder for models to learn patterns.

3. Cannot handle unseen phrases

Suppose training data contains:

very good
very bad

During inference:

extremely good

The bigram:

extremely good

was never seen before.

It gets ignored.

This is similar to the OOV (Out-of-Vocabulary) problem in BoW, but now it happens at the **phrase level**.

4. Loses long-range context

Consider:

"The movie that I watched yesterday was not good."

The important relationship is:

not → good

But they are far apart.

Bigram sees

not good

only if the words are adjacent.

✓

✓ If the sentence is:
"The movie was not really very good."

Bigrams become:

not really

really very

very good

The phrase "**not good**" is never captured.

5. Higher n means more data needed

Bigram

machine learning

may appear many times.

Trigram

machine learning model

appears less often.

4-gram

machine learning model training

appears even less.

As n increases:

- Features become rarer
- More training data is needed

6. Doesn't understand semantics

N-grams match exact words.

Training

good movie

Test

excellent movie

The model does not know:

good ≈ excellent

because they are different tokens.

N-grams capture order, but not semantic similarity.